

Photovoltaic Station Power Forecasting for Integrated Energy Systems Based on Spatiotemporal Features

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Abstract—With the increasing energy shortages and environmental degradation, the proportion of traditional fossil fuel-based power generation is gradually declining, being replaced by integrated energy systems that incorporate photovoltaic (PV) power generation, wind power generation, battery energy storage, and hydrogen-based power generation. However, PV power generation systems are highly influenced by environmental factors, exhibiting strong volatility and randomness, which creates challenges for grid dispatching. Conducting power forecasting for PV stations is thus essential to optimize grid operations. Yet, due to outdated communication systems in existing PV stations, integrated energy system cannot directly access operational data from all PV stations, making power forecasting difficult to implement. This paper proposes a method for predicting regional photovoltaic power generation using limited operational data from a few photovoltaic stations, based on long short-term memory (LSTM) network and graph neural networks (GNN). By deeply exploring the spatiotemporal characteristic relationships between these stations and the overall regional PV power generation, the method achieves data-driven, high-precision, and rapid power forecasting.

Keywords—*photovoltaic stations, spatiotemporal features, power forecasting*

I. INTRODUCTION

The global energy transition is being propelled by worsening energy shortages and environmental degradation, with integrated energy systems emerging as an effective solution. By rationally coordinating photovoltaic (PV) and wind power generation alongside battery and hydrogen energy storage, these systems achieve low-carbon, environmentally friendly, and sustainable practices across the entire energy supply chain—from production to consumption. Among these systems, photovoltaic (PV) stations have witnessed exponential growth in both deployment scale and capacity, emerging as a pivotal component of modern power grids [1]. However, the inherent intermittency of PV power generation—characterized by significant volatility and randomness due to meteorological dependencies—poses substantial challenges to grid stability and dispatching efficiency [2]. Accurate PV power forecasting has

thus become indispensable for optimizing grid operations, enabling proactive energy management and ancillary service allocation.

Despite its critical role, practical implementation of PV forecasting faces infrastructural barriers. Many existing PV stations, especially legacy installations, lack robust communication systems for real-time data transmission [3]. Consequently, grid operators cannot acquire operational data from all PV assets, rendering conventional data-intensive forecasting models ineffective. While recent studies have explored machine learning-based approaches for PV prediction [4], most methods assume comprehensive data availability, overlooking the real-world constraints of partial observability in regional PV clusters.

To address this gap, this paper proposes a spatiotemporal characteristic-driven framework for regional PV power forecasting using limited operational data from strategically sampled PV stations. The method establishes latent correlations between localized PV generation patterns and regional output through two key innovations: (1) PV stations with communication capabilities are selected based on Granger causality analysis to identify those most correlated with regional aggregate PV power generation. (2) A spatiotemporal characteristic-based regional PV power forecasting model is constructed. Validated against operational data from East China's PV fleet, our approach achieves a mean absolute percentage error (MAPE) of 0.62% under partial observability conditions. This work provides grid operators with a practical tool to mitigate forecasting uncertainties in regions with incomplete PV monitoring infrastructure.

The remainder of this paper is organized as follows: Section II discusses related work, Section III details the methodology, Section IV presents case studies, and Section V is conclusion.

II. RELATED WORK

The growing integration of PV stations into grids has intensified research on PV power forecasting, with methodologies evolving from physical models to data-driven

approaches. Traditional physical models, which rely on numerical weather prediction (NWP) and irradiance simulations, face limitations in handling the nonlinear relationships between meteorological variables and PV output [5]. Statistical methods, such as autoregressive integrated moving average (ARIMA), further struggle to capture the spatiotemporal dependencies inherent in distributed PV systems [6].

Data-driven forecasting models have gained prominence due to their ability to handle non-linear relationships between meteorological variables and power generation. Machine learning (ML) techniques, such as neural networks and support vector regression, are widely adopted for their adaptability to complex spatiotemporal patterns [4, 7]. For instance, hybrid models combining wavelet decomposition and deep learning have demonstrated improved accuracy in short-term predictions by capturing multi-scale features [8]. However, these methods typically rely on high-resolution operational data from individual PV stations, which may not be accessible in regions with outdated communication infrastructure [9].

Spatiotemporal correlation analysis has emerged as a key strategy to mitigate data scarcity. Studies like [10] emphasize the role of irradiance and weather similarity across neighboring PV plants in regional forecasting. Recent works propose clustering algorithms to identify geographically linked stations with analogous generation profiles, enabling partial data to infer regional trends [11]. For example, [8] introduced a hybrid model leveraging multi-frequency decomposition of sparse data to enhance prediction robustness.

Despite these advancements, challenges persist in scenarios where only limited operational data from a subset of PV stations is available. Traditional methods often overlook the latent spatiotemporal dependencies between localized generation and regional aggregation [12]. This gap highlights the need for frameworks that explicitly model these relationships without requiring full-grid data—a direction aligned with our proposed approach [13].

III. POWER FORECASTING MODEL BASED ON SPATIOTEMPORAL FEATURES

For geographically adjacent PV stations, their operational states exhibit similarities due to shared external environmental conditions. Therefore, the operational states of a small number of PV stations demonstrate a mapping relationship with the aggregate regional PV power generation. Concurrently, the temporal characteristics of individual PV stations inherently correlate across time series, allowing the prediction of future operational states through historical and current data. By considering the implicit spatiotemporal mapping relationships between current operational data and future power generation, it is feasible to perform PV power forecasting based on spatiotemporal features.

As shown in Fig.1, the proposed methodology proceeds as follows:

Granger Causality Analysis: For PV stations equipped with communication capabilities to acquire operational data, Granger causality analysis is applied to quantify inter-station correlations. Stations with strong causal relationships are grouped.

K-means Clustering: The grouped stations are further clustered using the K-means algorithm. Representative stations closest to cluster centroids are selected.

Spatiotemporal Prediction Model Training: Historical data from the representative stations trains a spatiotemporal feature-based power forecasting model.

Model Deployment: The trained model is deployed at the dispatch center to generate predictive power outputs.

Detailed explanations of each component are provided below:

A. Granger Causality Analysis for Inter-Station Correlation Quantification

Granger causality analysis is employed to statistically determine directional causal relationships between time-series data from geographically adjacent PV stations. Given two stations A and B , station A is said to "Granger-cause" B if historical data of A significantly improves the prediction of B 's future power output. This is formalized via vector autoregression (VAR) models:

$$B_t = \sum_{i=1}^p \alpha_i B_{t-i} + \sum_{i=1}^p \beta_i A_{t-i} + \varepsilon_t \quad (1)$$

where p is the lag order, and the null hypothesis ($\beta_i = 0$) is tested using F-statistics. Stations with F-statistic > 3.84 (95% confidence level) are grouped.

B. K-means Clustering for Representative Station Selection

All PV power stations are divided into N groups based on Granger causality analysis, with one representative station selected from each group. First, the generated power of each station is normalized using min-max scaling according to (2):

$$P_{\text{norm}_i} = \frac{P_i - P_{\min}}{P_{\max} - P_{\min}} \quad (2)$$

Where $P_{\max}$ and $P_{\min}$ represent the maximum and minimum power values of the station, respectively.

The K-means cluster centers for M PV stations within the same group across T time steps are calculated as shown in (3):

$$C_k = \frac{1}{M} \sum_{i=1}^M P'_{\text{norm}_i} \quad t \in \{1, 2, \dots, T\} \quad (3)$$

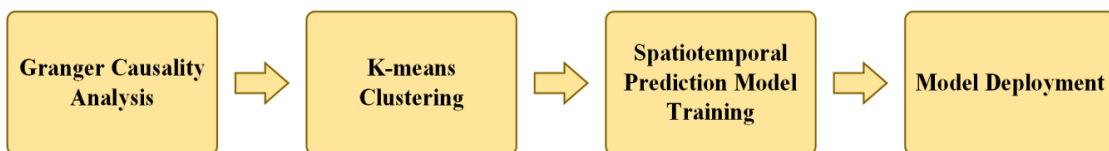


Fig. 1. Flowchart of the proposed methodology

The Euclidean distance from each PV station within the group to the cluster center over T time steps is calculated as shown in (4):

$$d_i = \sqrt{\sum_{t=1}^T (P_i^t - C_k^t)^2} \quad i \in \{1, 2, \dots, M\} \quad (4)$$

Select the PV station with the smallest Euclidean distance within each group as the representative station.

C. Spatiotemporal Feature-Based Forecasting Model Training

The operational data from the selected m groups of representative PV stations are used as the dataset to train the power forecasting model. As shown in Fig. 2, the model consists of three components: Long short-term memory (LSTM) network extracts temporal features from each selected PV station. Graph neural network (GNN) captures spatial features across all PV stations. Integrates temporal and spatial features through an artificial neural network (ANN) to output the predicted value for the next time step. The details are as follows:

LSTM networks are a specialized type of recurrent neural network (RNN) designed to model temporal dependencies in sequential data. Unlike vanilla RNNs, LSTMs address the vanishing gradient problem through gated mechanisms, enabling effective learning of both short-term and long-term patterns—a critical capability for PV power forecasting.

The LSTM controls how much past information is forgotten and how much is propagated to future states through:

$$\begin{cases} f_k = \sigma(W_f \cdot [h_{k-1}, X_k] + b_f) \\ i_k = \sigma(W_i \cdot [h_{k-1}, X_k] + b_i) \\ o_k = \sigma(W_o \cdot [h_{k-1}, X_k] + b_o) \end{cases} \quad (5)$$

Where, f_k , i_k and o_k represent the forget gate, input gate, and output gate, respectively. W_f , W_i and W_o are the weight matrices of the three gates. $[h_{k-1}, X_k]$ denotes the concatenation of the hidden state vector h_{k-1} and the input vector X_k . $\sigma(x)$ is the sigmoid function, which maps x to the interval (0, 1).

By capturing temporal features of the data through LSTM, the model can identify variation patterns across multiple time scales, such as sunrise-sunset cycles and building shadowing dynamics, thereby enabling more accurate prediction of PV station power output.

GNNs are a class of deep learning models designed to process data structured as graphs, where entities (nodes) and their relationships (edges) are explicitly modeled. GNNs are particularly effective for capturing spatial dependencies in non-Euclidean domains, such as power grids, social networks, or geographically distributed PV systems.

GNNs iteratively update node representations by aggregating information from neighboring nodes. The core operation is:

$$h_i^{l+1} = \phi \left(h_i^l, \bigoplus_{j \in N_i} \psi(h_i^l, h_j^l, e_{ij}) \right) \quad (6)$$

Where h_i^l is hidden state of node i at layer l , N_i is neighbors of node i .

In PV power forecasting, nodes represent individual PV stations, edges defined by spatial proximity and features include historical power output, irradiance, and weather conditions. GNN layers propagate and aggregate features across the graph, capturing regional irradiance patterns, cloud movement effects and grid topology constraints.

ANN is the most commonly used intelligent algorithm, capable of achieving accurate nonlinear mapping by adjusting the number of hidden layers and neurons. Since the feature values output by LSTM and GNN are of different dimensions and cannot be directly used, ANN is employed to perform the nonlinear mapping from spatiotemporal features to predicted data by treating each element in the matrix or array as an input neuron.

During model training, the data from selected representative PV stations are chronologically ordered and used as input, while the subsequent regional PV power output serves as the target. The residual network iteratively refines the model through feedback mechanisms until the accuracy plateaus or the

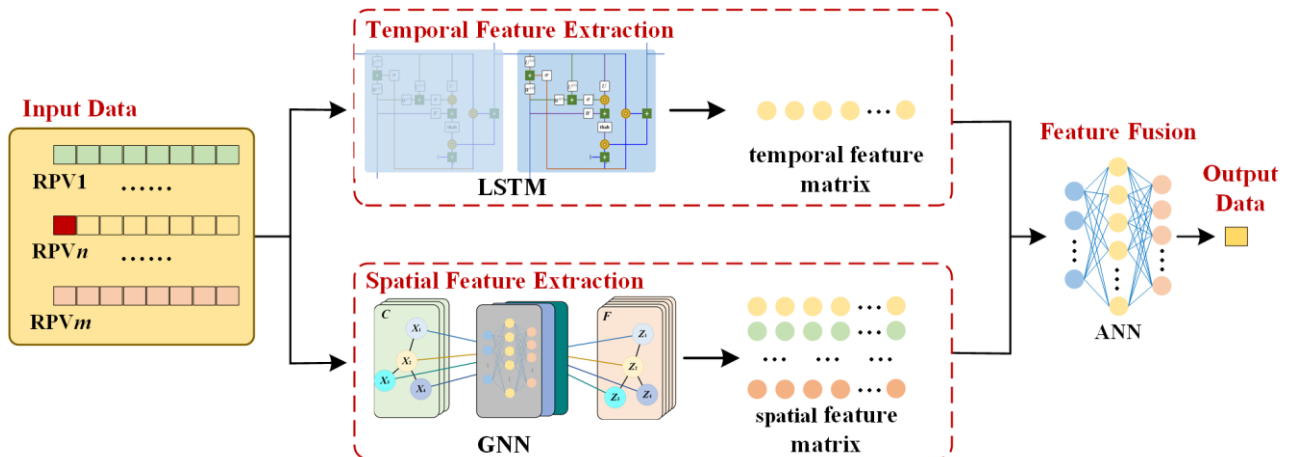


Fig. 2. Spatiotemporal feature-based PV forecasting model

maximum training epochs are reached, at which point training concludes.

The trained model is deployed at the dispatch center, which continuously receives real-time power data from representative PV stations. Based on these inputs, the model predicts regional PV power output for the next time step and thereby generates dispatch instructions to optimize the coordination of renewable energy systems, including wind, solar, storage, and hydrogen.

IV. CASE STUDY

This study selected 20 PV stations in a certain region of East China as research objects, forming a heterogeneous sample set with diversified characteristics. The selected plants exhibit significant variations in operational history (ranging from 5 to 9 years), installed capacity (distributed between 2MW to 30MW), and geographical distribution across 2 different micro-climate zones (plain and mountain). Notably, the temporal resolution of power data reporting from these plants varies among 1-minute, 2-minute, 5-minute, and 10-minute intervals. To harmonize the heterogeneous temporal datasets, we employed a greatest common divisor approach, establishing 10-minute intervals as the unified prediction time step through data aggregation and alignment techniques.

First, Granger causality analysis and K-means clustering were applied to select 6 representative PV stations from the group.

Subsequently, the 10-minute interval data from the six PV arrays were chronologically ordered, with one day constituting a single sample. This process constructed a dataset comprising 278 valid data entries, of which 178 entries were allocated to the training set and the remaining 100 entries reserved for the testing set.

To demonstrate the effectiveness and advancement of the method proposed in this paper, we employed commonly used Artificial Neural Networks (ANNs) and Convolutional Neural Networks (CNNs) as control groups for comparative testing. It should be noted that both the proposed method and the control

methods utilized identical training and testing datasets. Mean absolute percentage error (MAPE) and root mean square percentage error (RMSPE) were adopted as evaluation metrics, where lower values indicate closer proximity to the true results and thus higher prediction accuracy. The testing results are shown in Table I. Additionally, Fig. 3 illustrates the prediction outcomes of one set of testing data. Clearly, the proposed method demonstrates superior power forecasting performance.

V. CONCLUSION

This paper proposes a spatiotemporal feature-based power forecasting method for PV stations. By extracting spatiotemporal features from different PV stations within the same region, the method achieves high-precision regional PV power forecasting using a limited number of representative stations, thereby providing crucial support for grid dispatch and integrated energy system optimization. Experimental validation confirms that the proposed method demonstrates marked advancement compared to conventional forecasting models.

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TABLE I. ACCURACY FOR PROPOSED METHOD AND CONTROL METHODS

	MAPE	RMSPE
ANN	3.65	13.65
CNN	1.97	7.33
Proposed	0.62	2.15

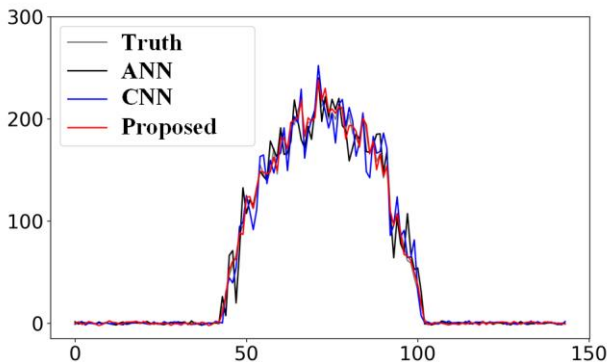


Fig. 3. Comparison of forecasting results for 3 methods